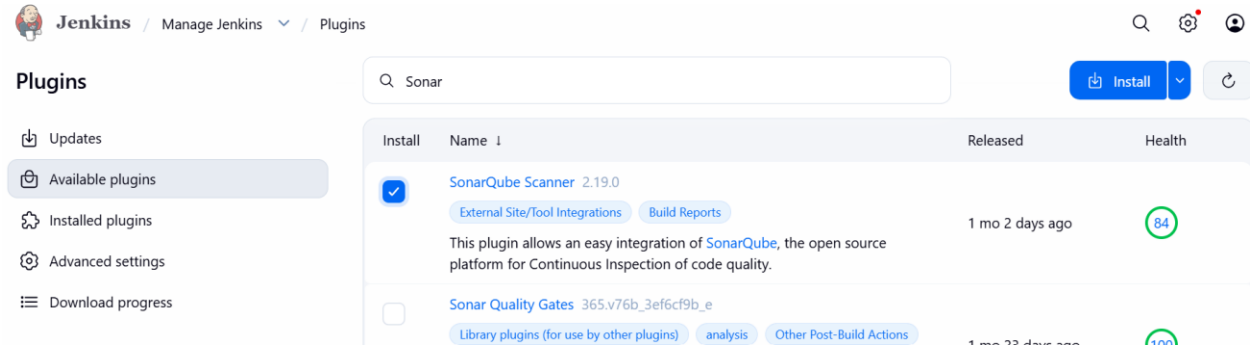
Create one Jenkins job using the below code and create three stages.

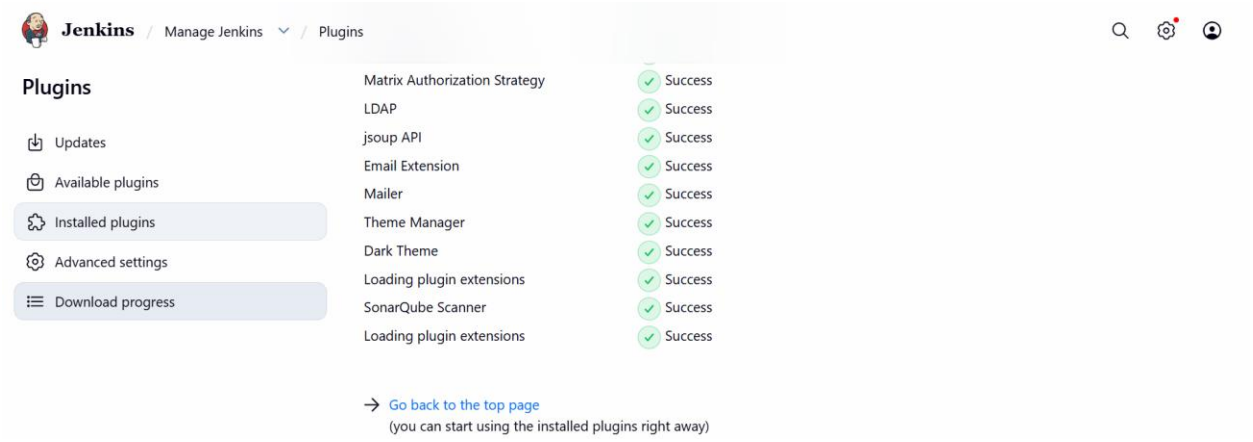
stage1: Git clone to download the source code.

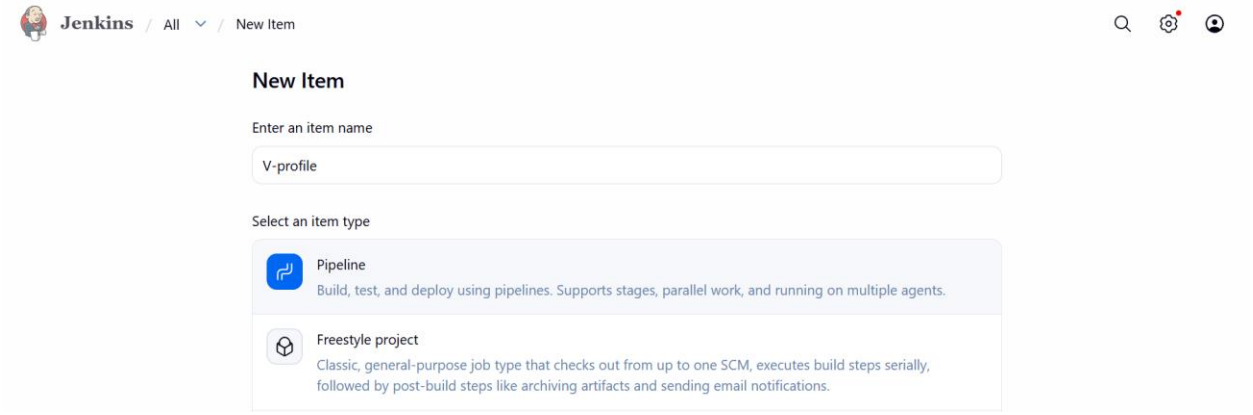
Create an instance and download java and git

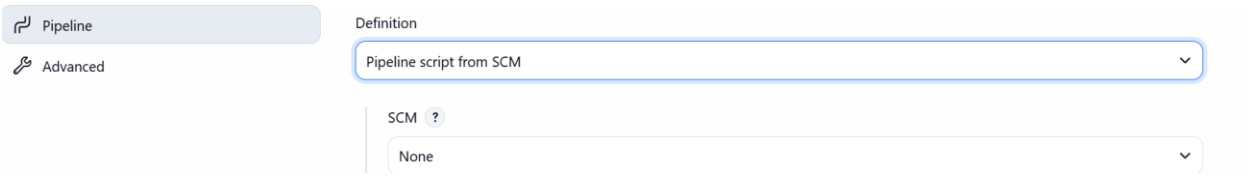
Then in the jenkins plugin search for Sonarqube scanner



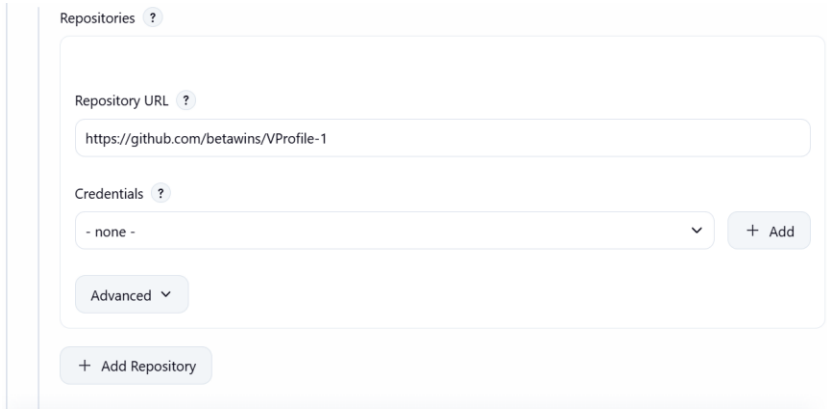
Install sonarqube scanner





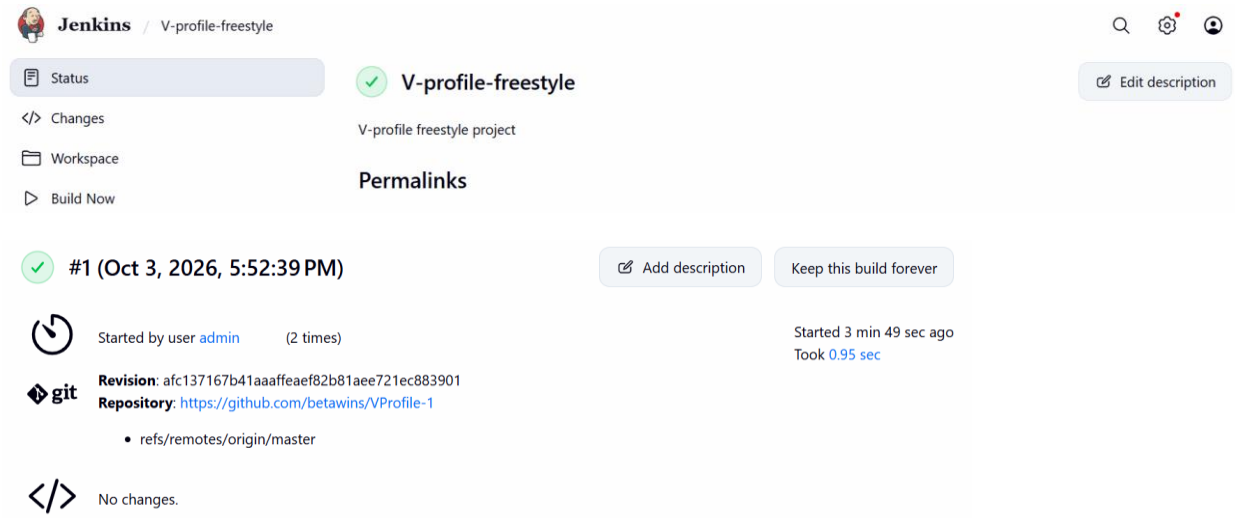


Add the repository url and it is public right, then no need of any credentials




```
        at java.base/java.nio.file.Files.readAllBytes(Files.java:3288)
        at java.base/java.nio.file.Files.readString(Files.java:3359)
        at hudson.FilePath$ReadToString.invoke(FilePath.java:2482)
        at hudson.FilePath$ReadToString.invoke(FilePath.java:2477)
        at hudson.FilePath.act(FilePath.java:1226)
        at hudson.FilePath.act(FilePath.java:1209)
        at hudson.FilePath.readToString(FilePath.java:2474)
        at PluginClassLoader for workflow-
cps//org.jenkinsci.plugins.workflow.cps.CpsScmFlowDefinition.create(CpsScmFlowDefinition.java:200)
        at PluginClassLoader for workflow-
cps//org.jenkinsci.plugins.workflow.cps.CpsScmFlowDefinition.create(CpsScmFlowDefinition.java:70)
        at PluginClassLoader for workflow-
job//org.jenkinsci.plugins.workflow.job.WorkflowRun.run(WorkflowRun.java:342)
        at hudson.model.ResourceController.execute(ResourceController.java:97)
        at hudson.model.Executor.run(Executor.java:456)
Finished: FAILURE
```

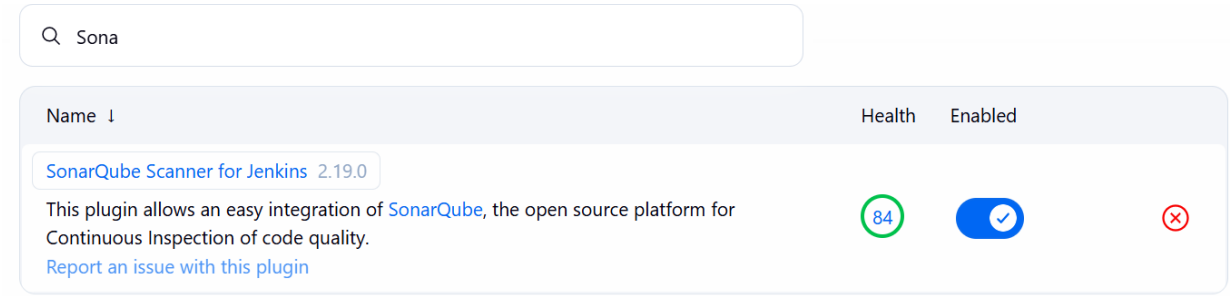
The error is because of the low disc space



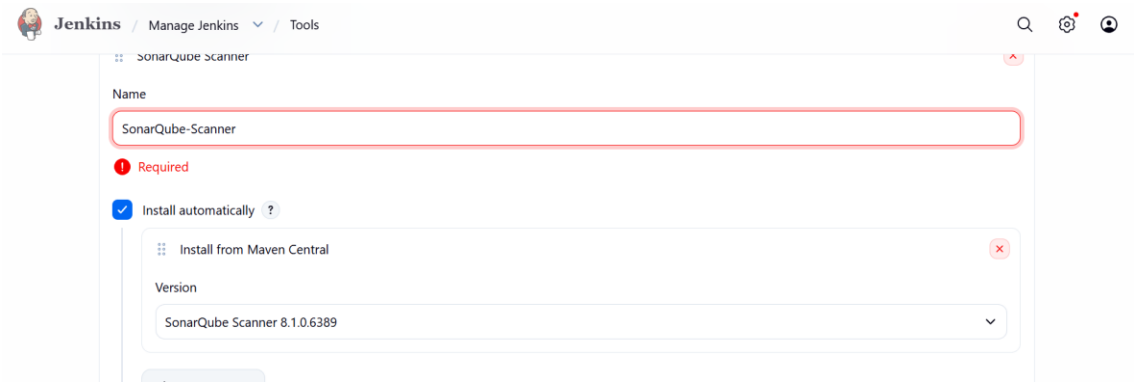
```
[root@ip-10-0-15-30 ~]# git --version
git version 2.50.1
[root@ip-10-0-15-30 ~]# ls
[root@ip-10-0-15-30 ~]# cd /var/lib/jenkins/
[root@ip-10-0-15-30 jenkins]# ls
caches                                          identity.key.enc                                nodeMonitors.xml          updates
config.xml                                      jenkins.install.InstallUtil.lastExecVersion     plugins                   userContent
hudson.model.UpdateCenter.xml                   jenkins.install.UpgradeWizard.state             queue.xml                 users
hudson.plugins.emailext.ExtendedEmailPublisher.xml  jenkins.model.JenkinsLocationConfiguration.xml  secret.key                workspace
hudson.plugins.git.GitTool.xml                  jenkins.telemetry.Correlator.xml                secret.key.not-so-secret
hudson.plugins.sonar.SonarGlobalConfiguration.xml  jobs                                         secrets
[root@ip-10-0-15-30 jenkins]# cd workspace
[root@ip-10-0-15-30 workspace]# ls
V-profile-freestyle  V-profile@script
[root@ip-10-0-15-30 workspace]# cd V-profile-freestyle
[root@ip-10-0-15-30 V-profile-freestyle]# ls
Dockerfile  README.md  ansible  azure-pipelines.yml  pom.xml  src
[root@ip-10-0-15-30 V-profile-freestyle]#
```

Git clone done

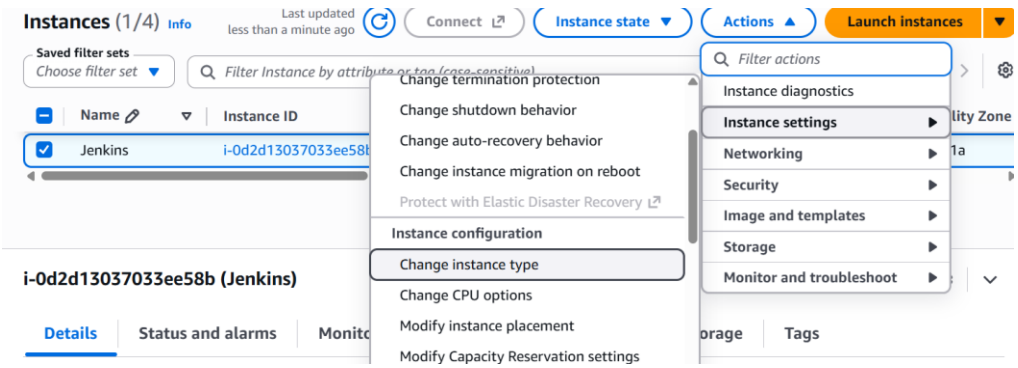
stage2: Sonarqube Integration to check the quality of code.

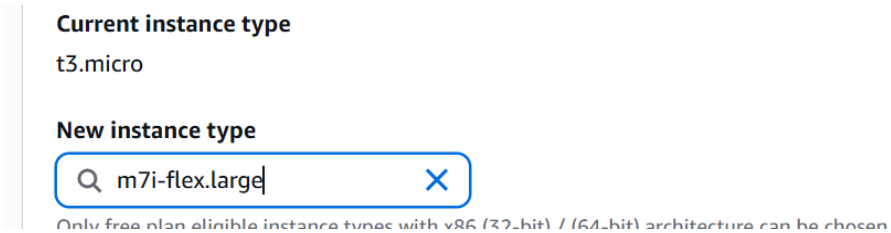


Sonarqube scanner already installed



For sonarqube instanve type should be large so,






```
[root@ip-10-0-2-103 ~]# wget https://binaries.sonarsource.com/Distribution/sonarqube/sonarqube-26.9.0.129388.zip
```

Now install the sonarqube

And unzip that

```
[root@ip-10-0-2-103 ~]# ls
sonarqube-26.9.0.129388  sonarqube-26.9.0.129388.zip
[root@ip-10-0-2-103 ~]# mv sonarqube-26.9.0.129388 sonarqube
[root@ip-10-0-2-103 ~]# ls
sonarqube  sonarqube-26.9.0.129388.zip
[root@ip-10-0-2-103 ~]#
```

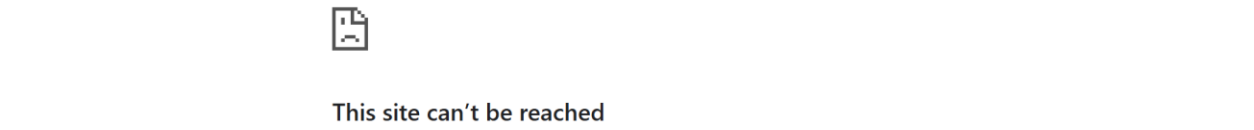
```
[root@ip-10-0-2-103 ~]# useradd -m -s /bin/bash sonar
[root@ip-10-0-2-103 ~]# chown -R sonar:sonar /sonarqube
chown: cannot access '/sonarqube': No such file or directory
[root@ip-10-0-2-103 ~]# chown -R sonar:sonar /root/sonarqube
[root@ip-10-0-2-103 ~]# mv /root/sonarqube /opt/sonarqube
[root@ip-10-0-2-103 ~]# ls -ld /opt/sonarqube
drwxr-xr-x. 14 sonar sonar 16384 Sep  2 09:48 /opt/sonarqube
[root@ip-10-0-2-103 ~]# su -sonar
su: failed to execute onar: No such file or directory
[root@ip-10-0-2-103 ~]# which bash
/usr/bin/bash
[root@ip-10-0-2-103 ~]# usermod -s /usr/bin/bash sonar
[root@ip-10-0-2-103 ~]# su -sonar
su: failed to execute onar: No such file or directory
[root@ip-10-0-2-103 ~]# su - sonar
[sonar@ip-10-0-2-103 ~]$ cd /opt/sonarqube/bin/linux-x86-64
[sonar@ip-10-0-2-103 linux-x86-64]$ ./sonar.sh start
-bash: ./sonar.sh: Permission denied
[sonar@ip-10-0-2-103 linux-x86-64]$ ls -l sonar.sh
```

```
[sonar@ip-10-0-2-103 linux-x86-64]$ ls -l sonar.sh
-rw-r--r--. 1 sonar sonar 7225 Sep  2 09:41 sonar.sh
[sonar@ip-10-0-2-103 linux-x86-64]$ exit
logout
[root@ip-10-0-2-103 ~]# chmod +x /opt/sonarqube/bin/linux-x86-64/sonar.sh
[root@ip-10-0-2-103 ~]# su - sonar
Last login: Sat Oct  3 18:47:49 UTC 2026 on pts/1
[sonar@ip-10-0-2-103 ~]$ cd /opt/sonarqube/bin/linux-x86-64
[sonar@ip-10-0-2-103 linux-x86-64]$ ./sonar.sh start
/usr/bin/java
Starting SonarQube...
Started SonarQube.
[sonar@ip-10-0-2-103 linux-x86-64]$
```

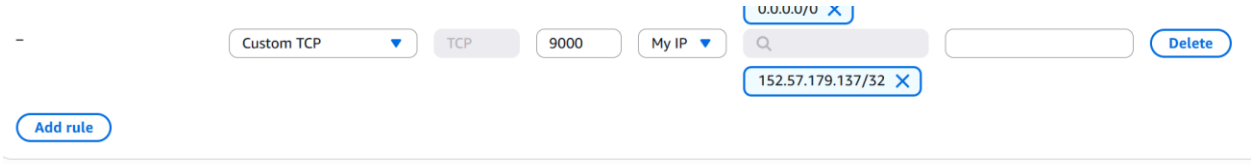
```
[sonar@ip-10-0-15-30 linux-x86-64]$ ./sonar.sh status
/usr/bin/java
SonarQube is running (3439).
[sonar@ip-10-0-15-30 linux-x86-64]$ curl http://localhost:9000
<!doctype html>
<html lang="en">
  <head>
    <script type="module" crossorigin src="/js/polyfills-CAZw4urH.js"></script>

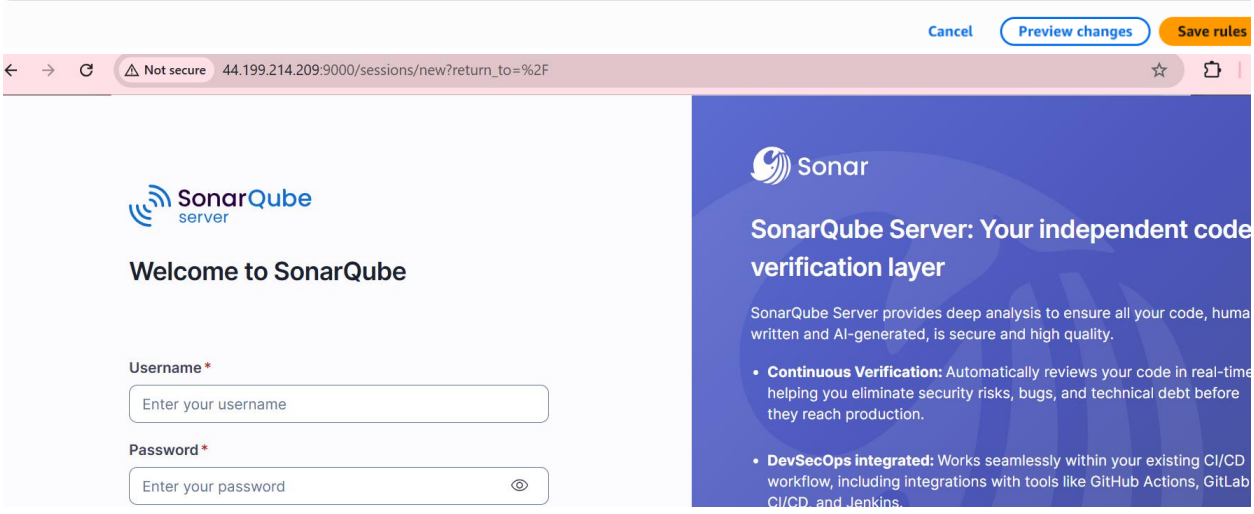
    <meta http-equiv="content-type" content="text/html; charset=UTF-8" charset="UTF-8" />
    <meta http-equiv="X-UA-Compatible" content="IE=edge" />
    <link rel="apple-touch-icon" href="/apple-touch-icon.png" />
    <link rel="icon" type="image/x-icon" href="/favicon.ico" />
    <meta name="application-name" content="SonarQube" />
    <meta name="msapplication-TileColor" content="#FFFFFF" />
    <meta name="msapplication-TileImage" content="/mstile-512x512.png" />
    <title>SonarQube</title>
    <script>
      window.__assetsPath = function (filename) {
        return '/' + filename;
      };
    </script>
```

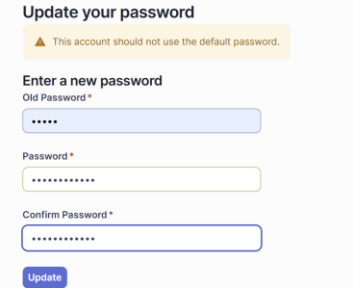




Now add the inbound rule in security groups



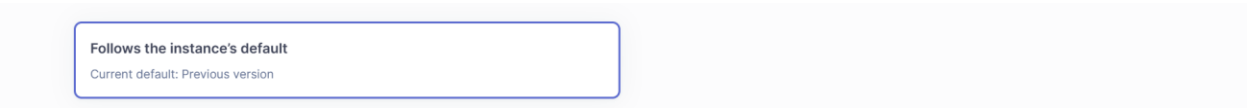




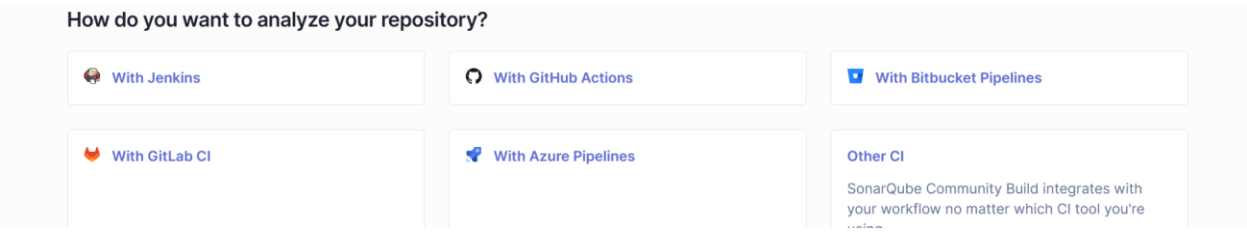
Update the password

After that click on the create the project manually

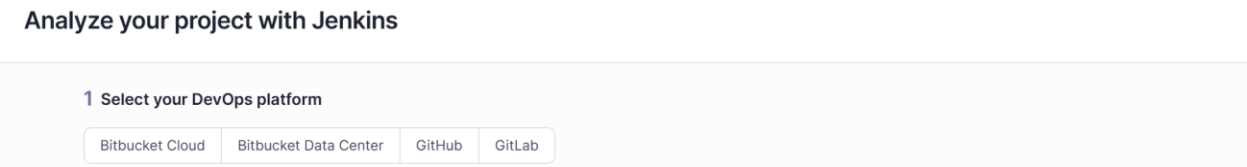




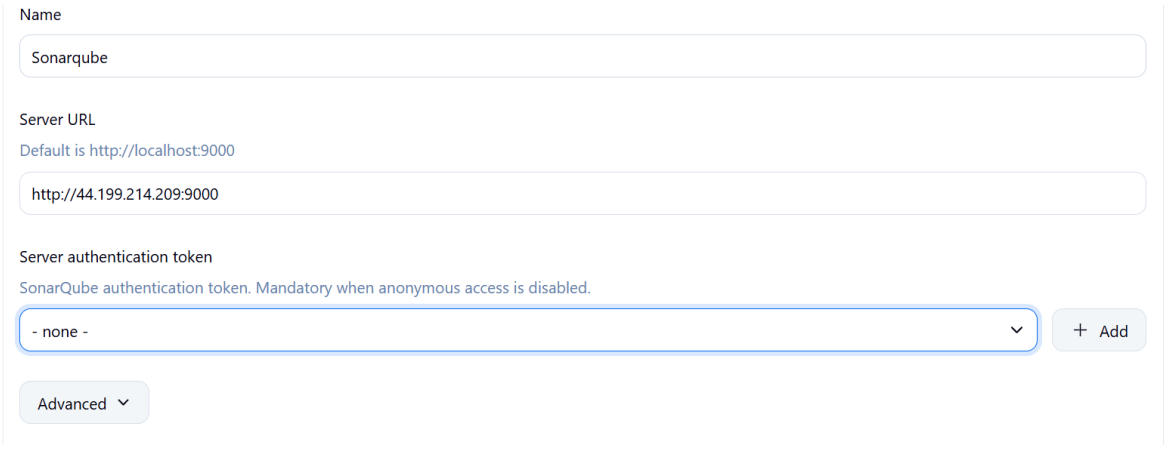
Then create project



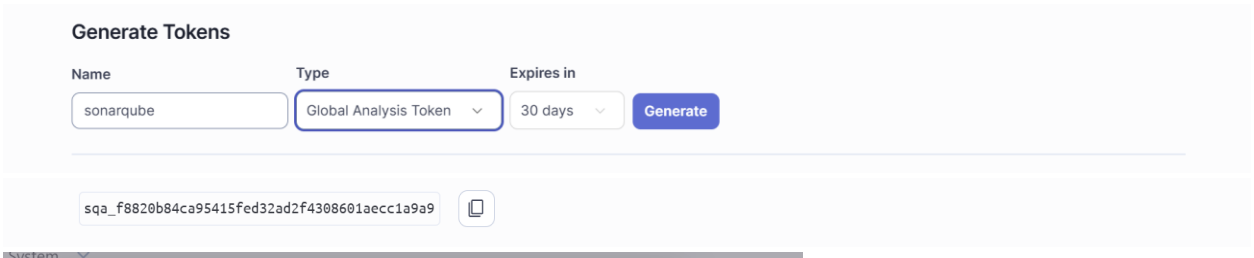
Choose jenkins

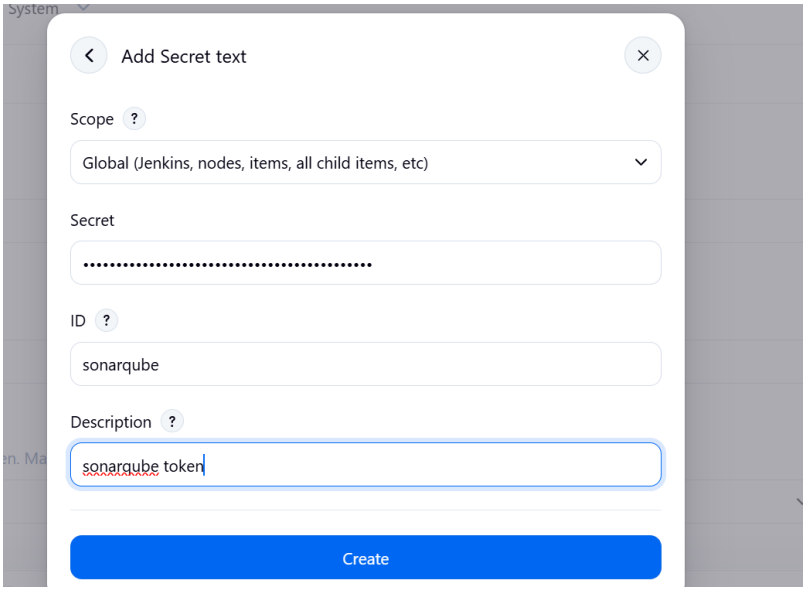


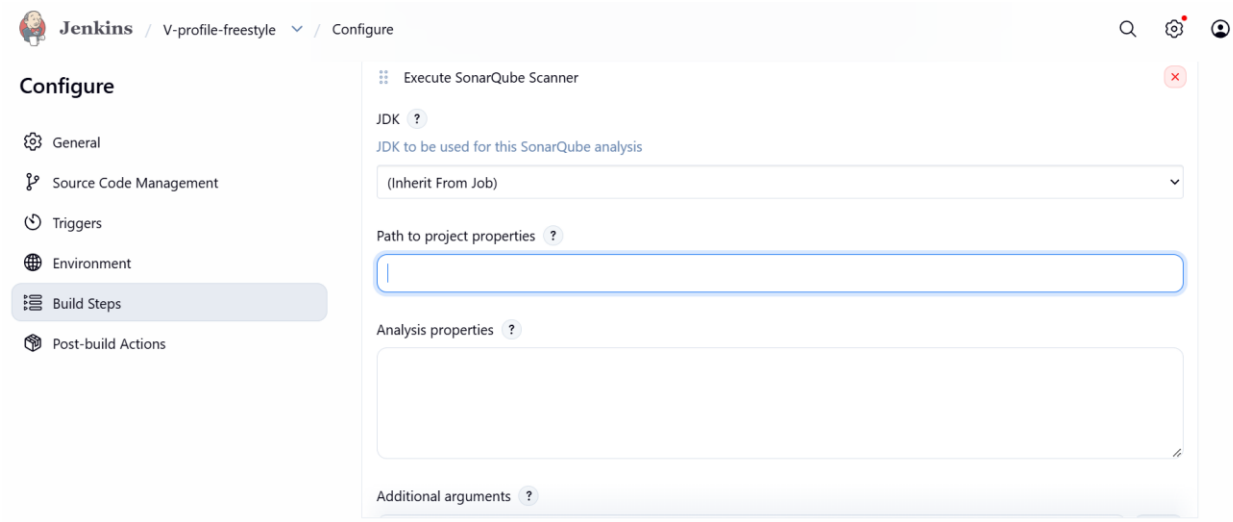
Click on the github



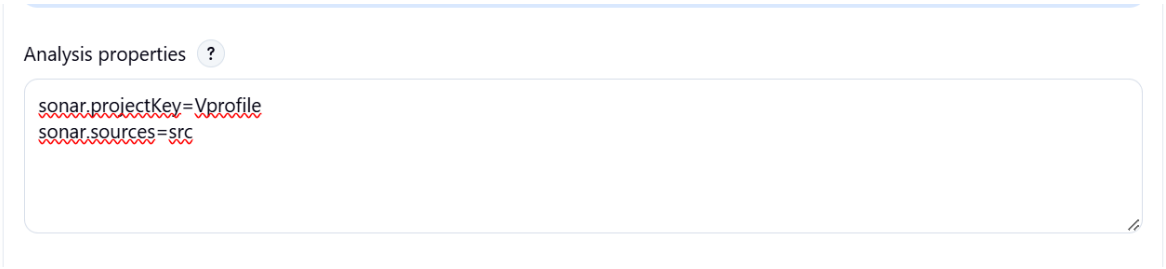
Now we need token, to securely autenticate with sonarqube



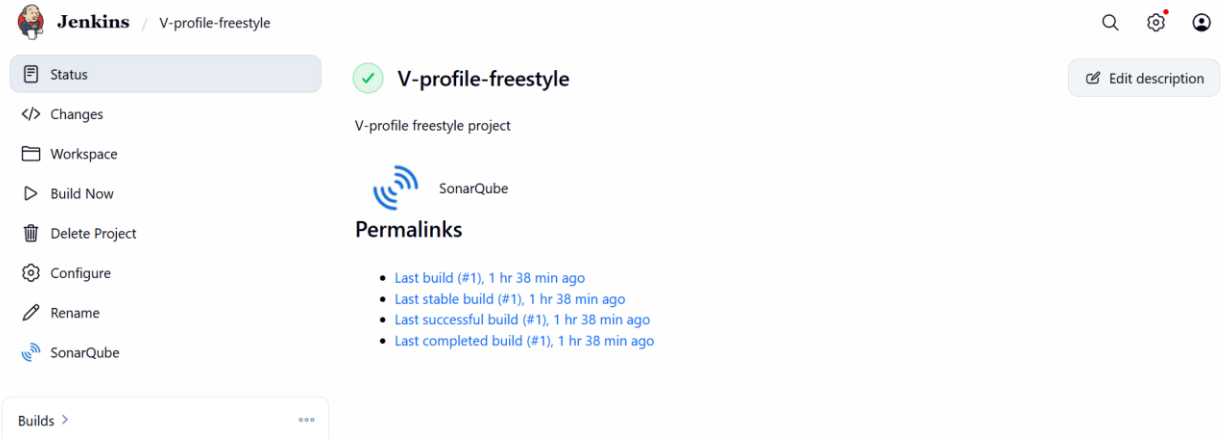




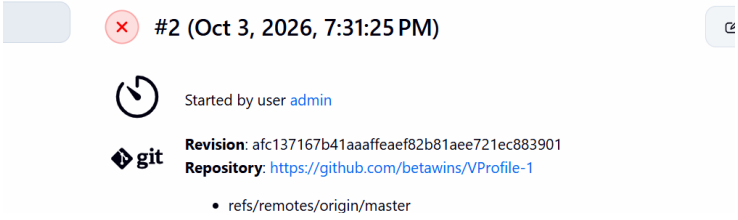
Add the job in jenkins

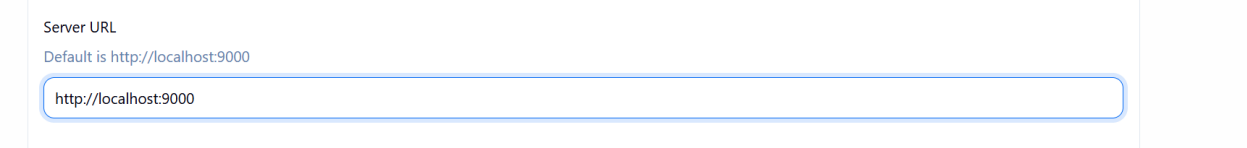


Then save it



Now our jenkins is integrated with the sonarqube





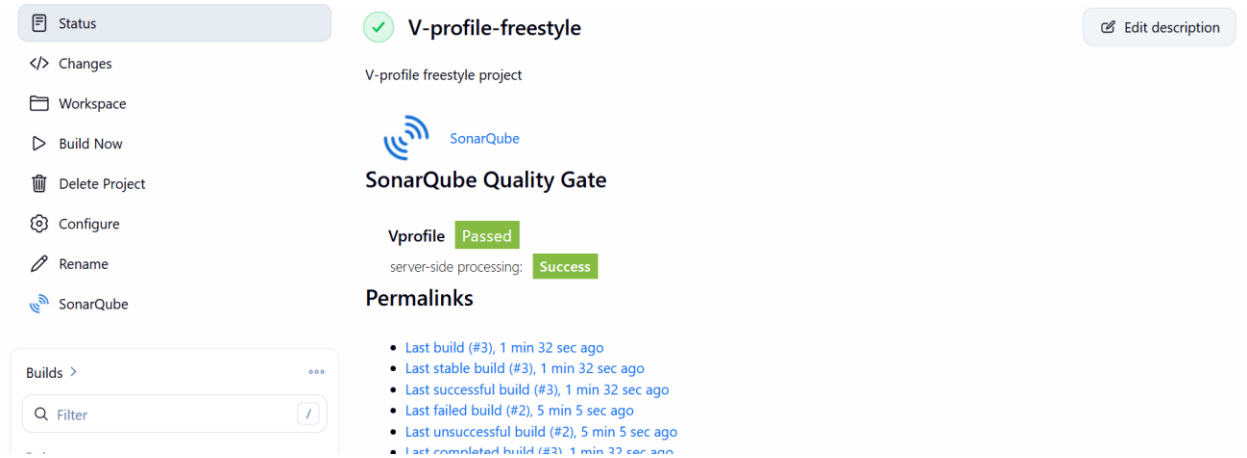
HTTP connect timed out while Jenkins is trying to reach:

http://44.199.214.209:9000

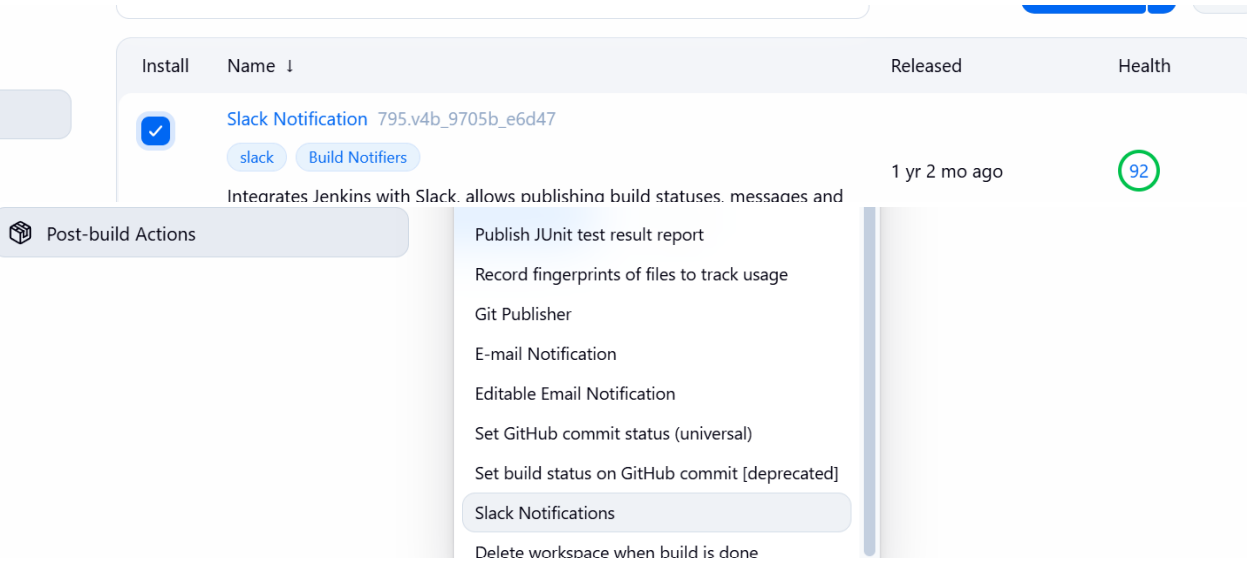
Since Jenkins and SonarQube are running on the same EC2 instance, Jenkins should connect to SonarQube through localhost, not the public IP.

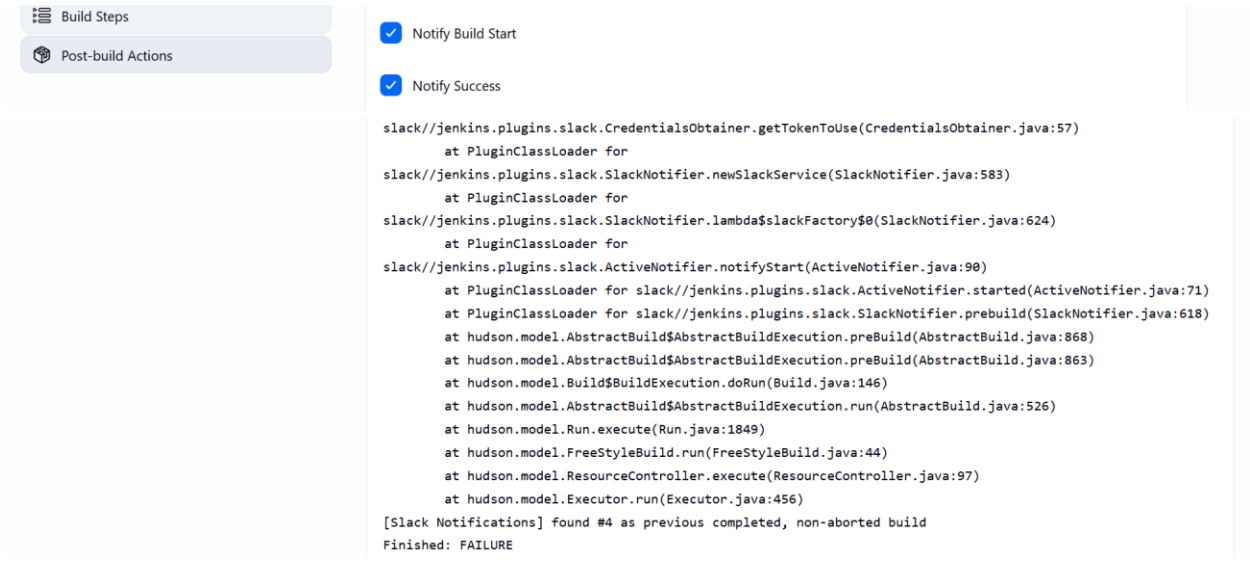
```
19:35:19.855 INFO  Total time: 19.776s
Finished: SUCCESS
```

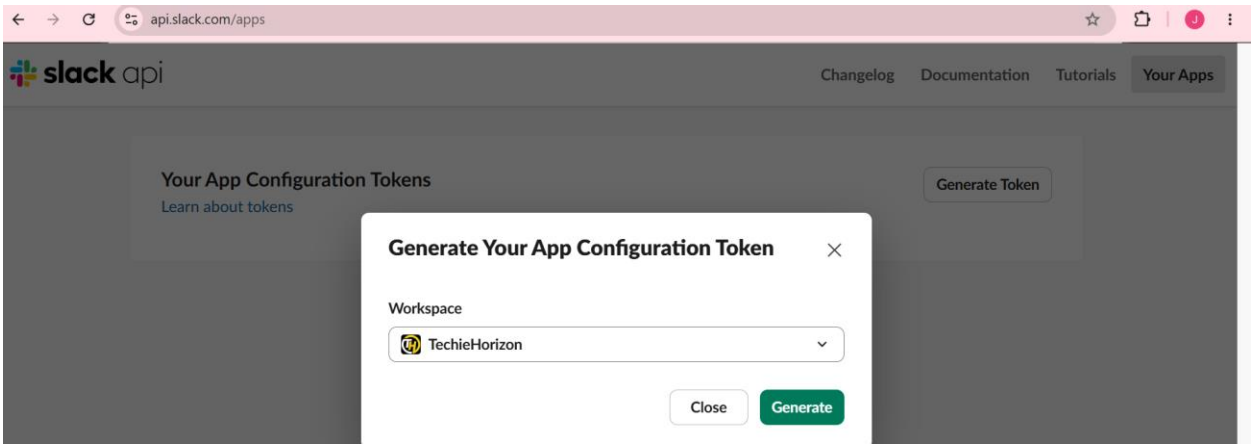
It is success now

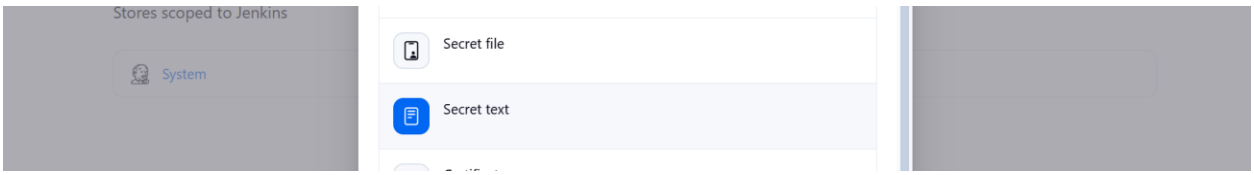


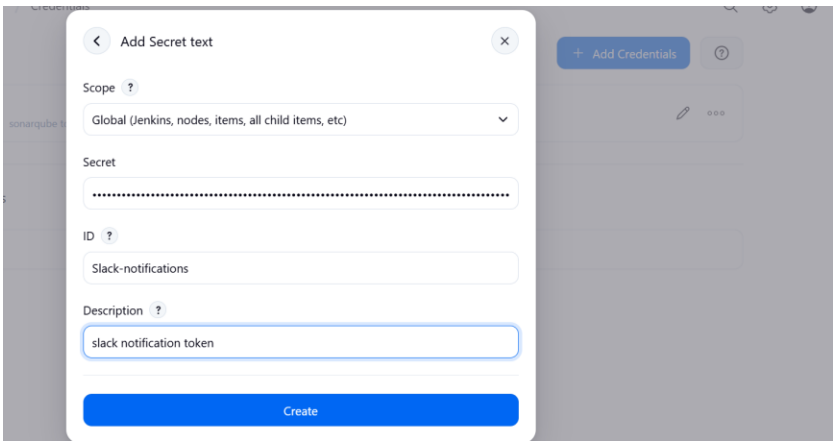
Install slack notification

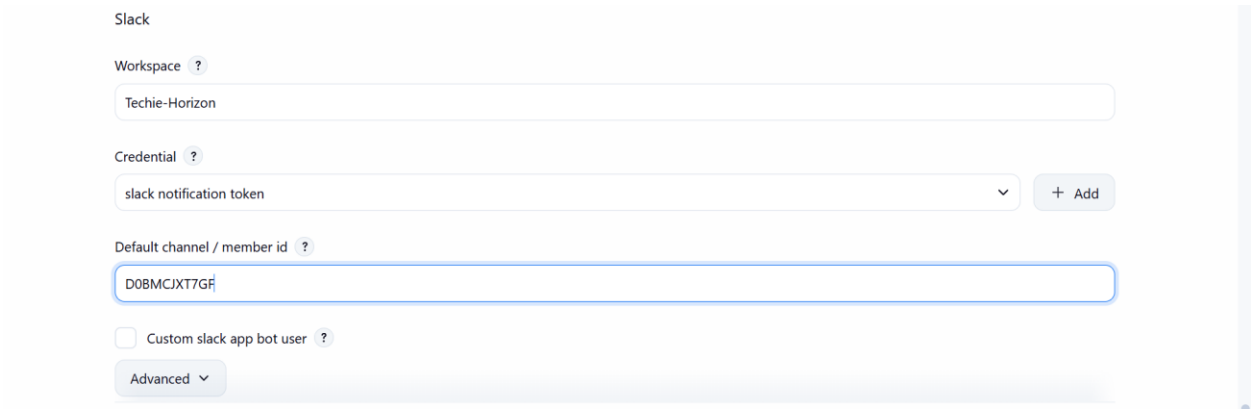


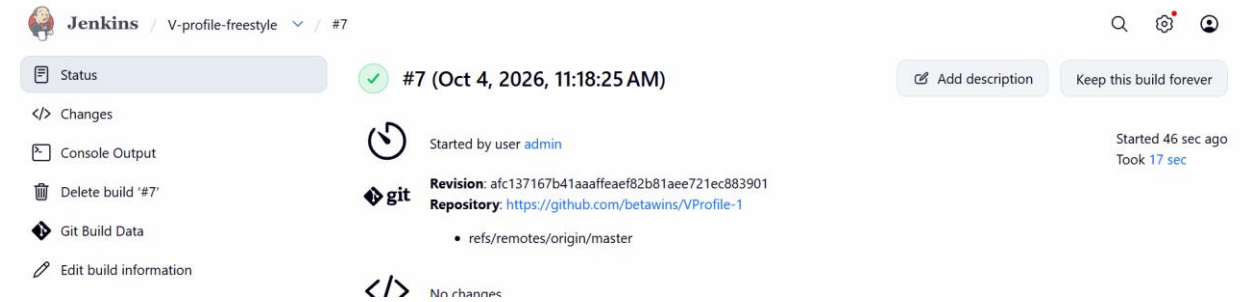

Build Steps
Post-build Actions
Notify Build Start
Notify Success
slack//jenkins.plugins.slack.CredentialsObtainer.getTokenToUse(CredentialsObtainer.java:57)
at PluginClassLoader for
slack//jenkins.plugins.slack.SlackNotifier.newSlackService(SlackNotifier.java:583)
at PluginClassLoader for
slack//jenkins.plugins.slack.SlackNotifier.lambda$slackFactory$0(SlackNotifier.java:624)
at PluginClassLoader for
slack//jenkins.plugins.slack.ActiveNotifier.notifyStart(ActiveNotifier.java:90)
at PluginClassLoader for slack//jenkins.plugins.slack.ActiveNotifier.started(ActiveNotifier.java:71)
at PluginClassLoader for slack//jenkins.plugins.slack.SlackNotifier.prebuild(SlackNotifier.java:618)
at hudson.model.AbstractBuild$AbstractBuildExecution.preBuild(AbstractBuild.java:868)
at hudson.model.AbstractBuild$AbstractBuildExecution.preBuild(AbstractBuild.java:863)
at hudson.model.Build$BuildExecution.doRun(Build.java:146)
at hudson.model.AbstractBuild$AbstractBuildExecution.run(AbstractBuild.java:526)
at hudson.model.Run.execute(Run.java:1849)
at hudson.model.FreeStyleBuild.run(FreeStyleBuild.java:44)
at hudson.model.ResourceController.execute(ResourceController.java:97)
at hudson.model.Executor.run(Executor.java:456)
[Slack Notifications] found #4 as previous completed, non-aborted build
Finished: FAILURE


api.slack.com/apps
slack api
Changelog
Documentation
Tutorials
Your Apps
Your App Configuration Tokens
Learn about tokens
Generate Token
Generate Your App Configuration Token
Workspace
TechieHorizon
Close
Generate

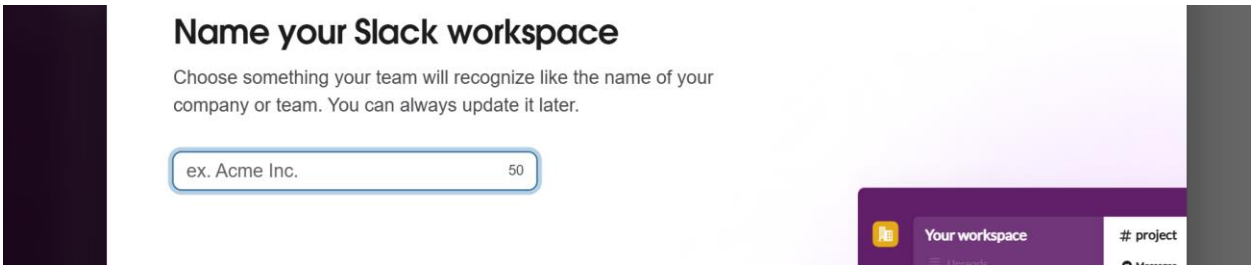

Stores scoped to Jenkins
System
Secret file
Secret text

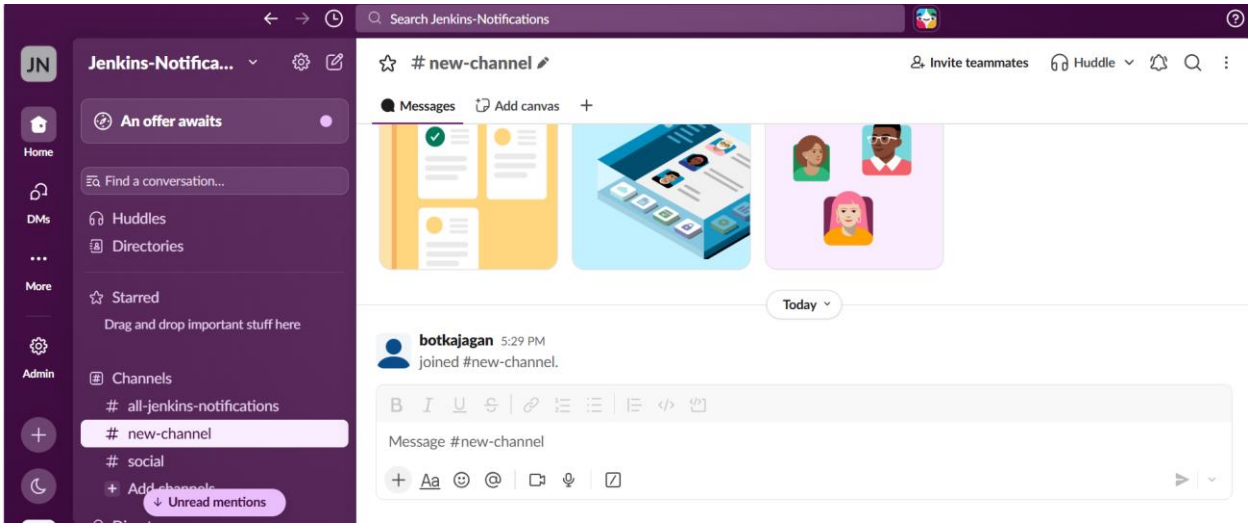

Add Secret text
Scope
Global (Jenkins, nodes, items, all child items, etc)
Secret
ID
Slack-notifications
Description
slack notification token
Create
Add Credentials

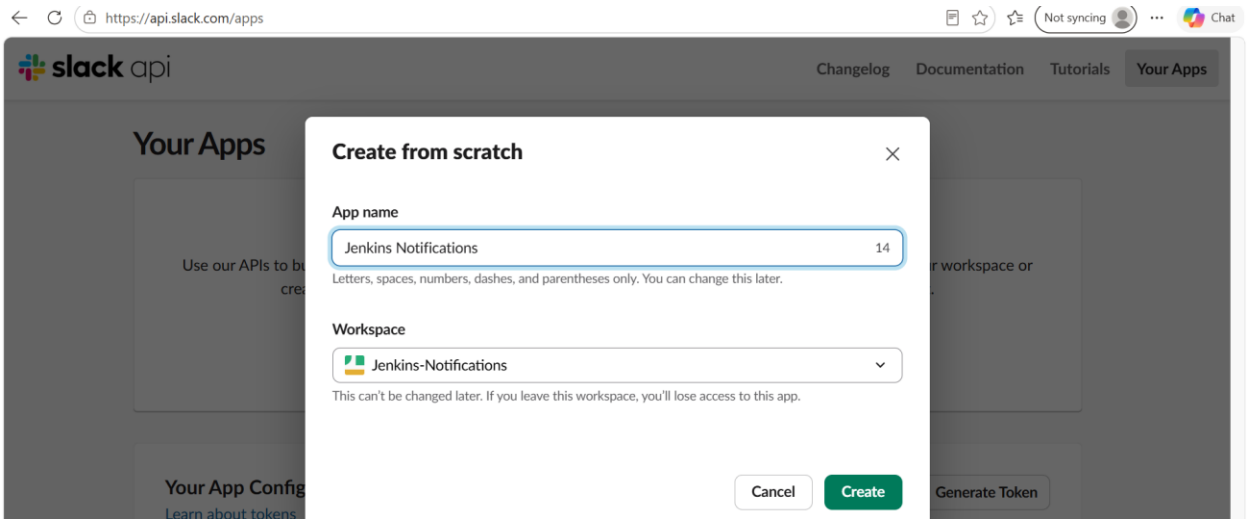

Slack
Workspace
Techie-Horizon
Credential
slack notification token
Add
Default channel / member id
D0BMCJXT7GF
Custom slack app bot user
Advanced


Jenkins
V-profile-freestyle
#7
Status
Changes
Console Output
Delete build '#7'
Git Build Data
Edit build information
#7 (Oct 4, 2026, 11:18:25 AM)
Add description
Keep this build forever
Started by user admin
Started 46 sec ago
Took 17 sec
Revision: afc137167b41aaaffeaef82b81aee721ec883901
Repository: https://github.com/betawins/VProfile-1
refs/remotes/origin/master
No changes

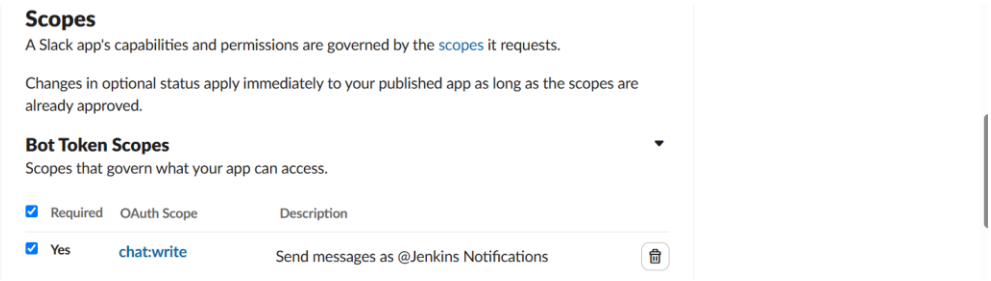

Add a workspace
Sign in to another workspace
Find workspaces
Create a new workspace
agents yet



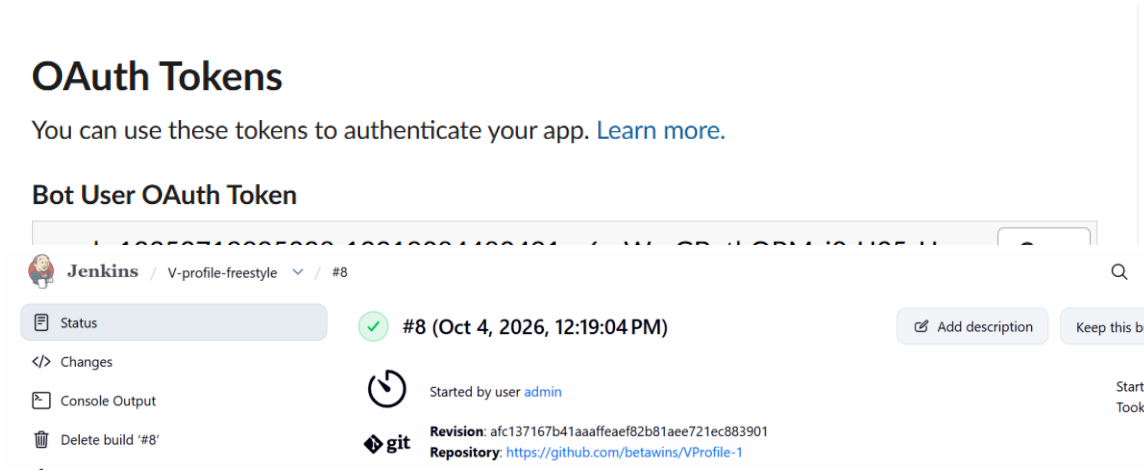




In the Oauth and permissions, choose



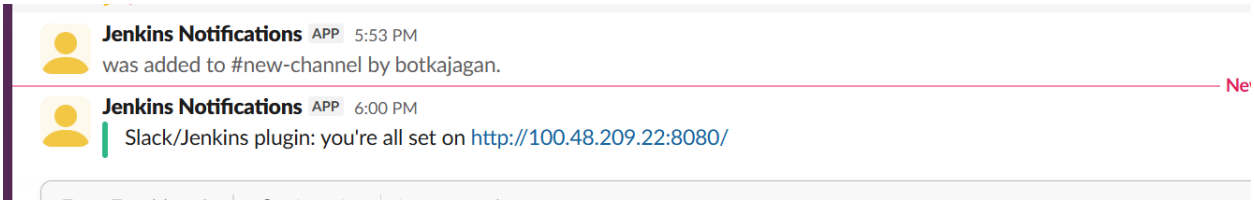
Next, click on to jenkins-Notification



this time the output is success

In the new channel message " /invite @Jenkins Notifications

```
[Slack Notifications] found #8 as previous completed, non-aborted build
[Slack Notifications] will send OnSuccessNotification because build matches and user preferences allow it
Finished: SUCCESS
```



Slack Integration is working now

Create one Jenkins job using the below code and create three stages.

stage1: Git clone to download the source code.

```
[root@ip-10-0-14-116 ~]# java --version
openjdk 21.0.12.1 2026-08-18 LTS
OpenJDK Runtime Environment Corretto-21.0.12.9.1 (build 21.0.12.1+9-LTS)
OpenJDK 64-Bit Server VM Corretto-21.0.12.9.1 (build 21.0.12.1+9-LTS, mixed mode, sharing)
```

Install Java

```
Installed:
  container-selinux-4:2.245.0-1.amzn2023.noarch          containerd-2.2.7-1.amzn2023.0.1.x86_64          docker-25.0.16-1.amzn2023.0.4.x86_64
  iptables-libs-1.8.8-3.amzn2023.0.2.x86_64              iptables-nft-1.8.8-3.amzn2023.0.2.x86_64        libcgroup-3.0-1.amzn2023.0.1.x86_64
  libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64     libnfnetlink-1.0.1-19.amzn2023.0.2.x86_64       libnftnl-1.2.2-2.amzn2023.0.2.x86_64
  pigz-2.5-1.amzn2023.0.4.x86_64                         runc-1.3.6-1.amzn2023.0.1.x86_64

Complete!
[root@ip-10-0-14-116 ~]#
```

Install docker

```
[root@ip-10-0-14-116 ~]# systemctl start docker
[root@ip-10-0-14-116 ~]# systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
[root@ip-10-0-14-116 ~]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-10-0-14-116 ~]# systemctl status docker
● docker.service - Docker Application Container Engine
     Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: disabled)
     Active: active (running) since Sun 2026-10-04 16:00:27 UTC; 1min 9s ago
TriggeredBy: ● docker.socket
```

```
[root@ip-10-0-14-116 ~]# docker IMAGE
docker: 'IMAGE' is not a docker command.
See 'docker --help'
[root@ip-10-0-14-116 ~]# docker images
REPOSITORY   TAG       IMAGE ID   CREATED   SIZE
```

```
[root@ip-10-0-14-116 ~]# docker pull jenkins/jenkins:lts-jdk21
lts-jdk21: Pulling from jenkins/jenkins
6eefb2f5d3e9: Pull complete
408777b27aa9: Pull complete
810008851f3a: Pull complete
d89089c8a5a3: Pull complete
701ded2f9b33: Pull complete
968c36a3b2c5: Pull complete
3dd4a99993ab: Pull complete
a20a14499a3e: Pull complete
463a574793f8: Pull complete
f9e31fab50e8: Pull complete
95bc7464aaea: Pull complete
a91427fe0e3f: Pull complete
Digest: sha256:a660310e39ade10631f774bacd5219de767dfd08947ea5c32a739b5e5bb382c1
Status: Downloaded newer image for jenkins/jenkins:lts-jdk21
docker.io/jenkins/jenkins:lts-jdk21
[root@ip-10-0-14-116 ~]#
```

It means we're going to download the Jenkins Docker image from Docker Hub onto your EC2.

Think of it like this:

Docker = the software that runs containers.

Jenkins image = a ready-made Jenkins package.

docker pull = download that package.

jenkins/jenkins:lts-jdk21 = Jenkins Long-Term Support version with Java 21.

```
[root@ip-10-0-14-116 ~]# systemctl stop jenkins
```

Because Jenkins was already installed directly on this EC2, we should stop it before starting Jenkins in Docker (otherwise port 8080 can conflict).

```
[root@ip-10-0-14-116 ~]# docker run -d --name jenkins -p 8080:8080 -p 50000:50000 jenkins/jenkins:lts-jdk21
05fa69c242dc6776408f281e25d21c598d9a4dec6cec5b52bf3da9913fe480be
[root@ip-10-0-14-116 ~]#
```

docker run → create and start a container

-d → run it in the background

--name jenkins → name the container jenkins

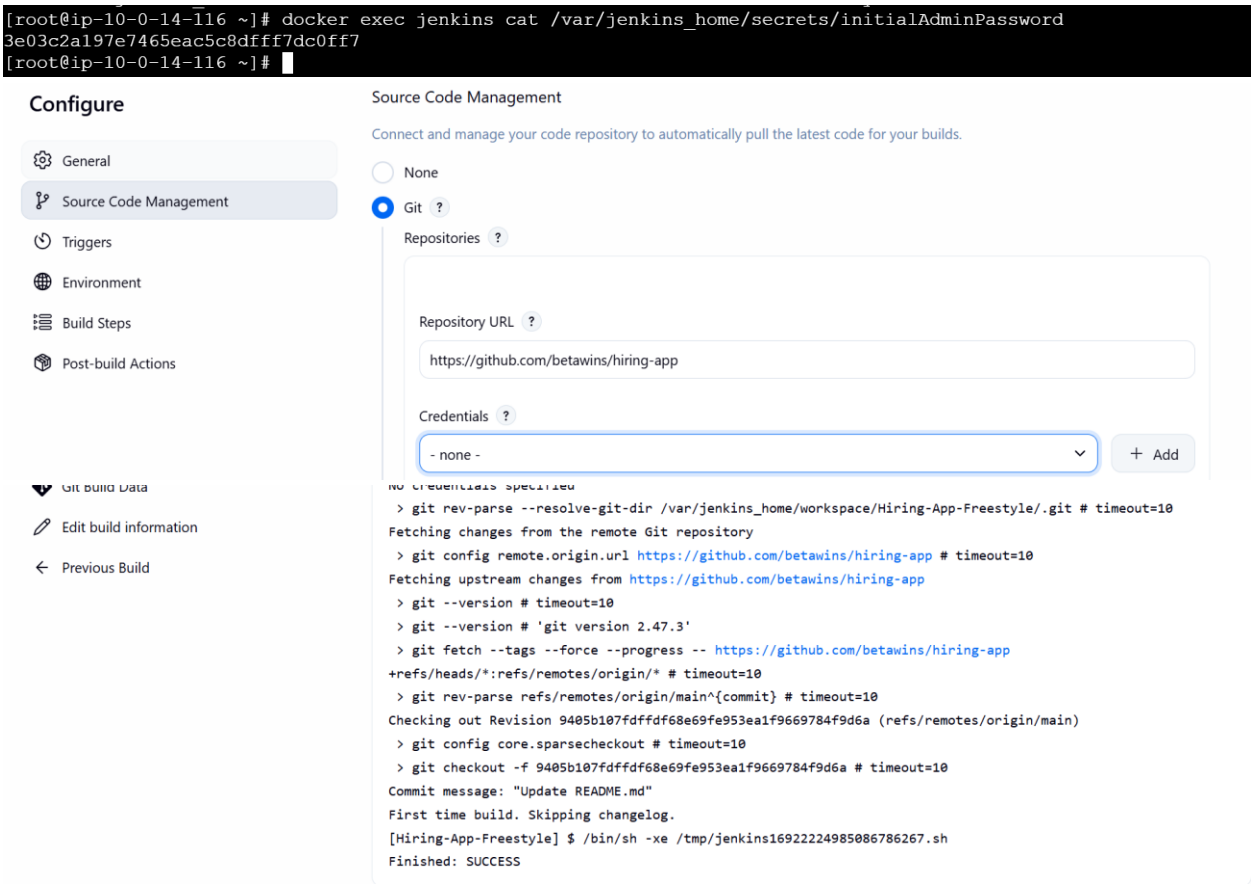
-p 8080:8080 → EC2 port 8080 → Jenkins port 8080

-p 50000:50000 → Jenkins agent communication port

jenkins/jenkins:lts-jdk21 → the Jenkins image we just downloaded

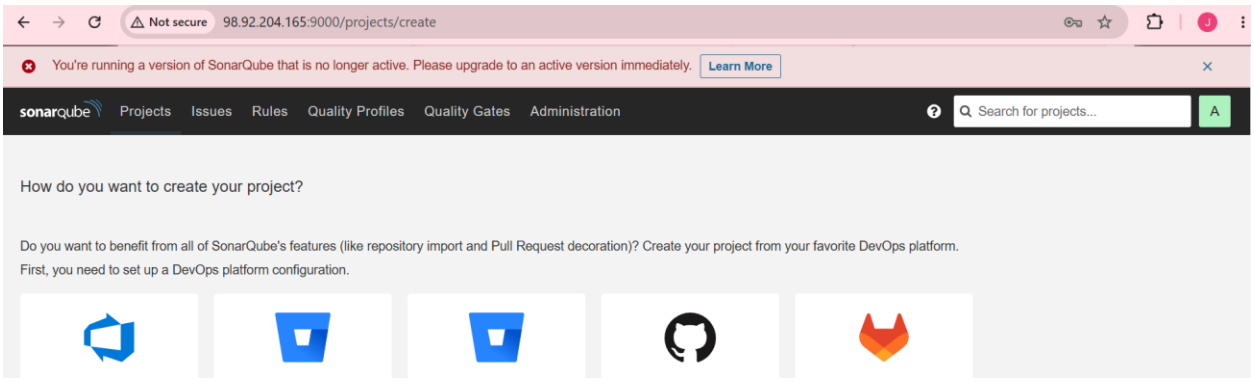
So after this command, Jenkins will be running as a Docker container, and we'll access it through:

http://<EC2-Public-IP>:8080

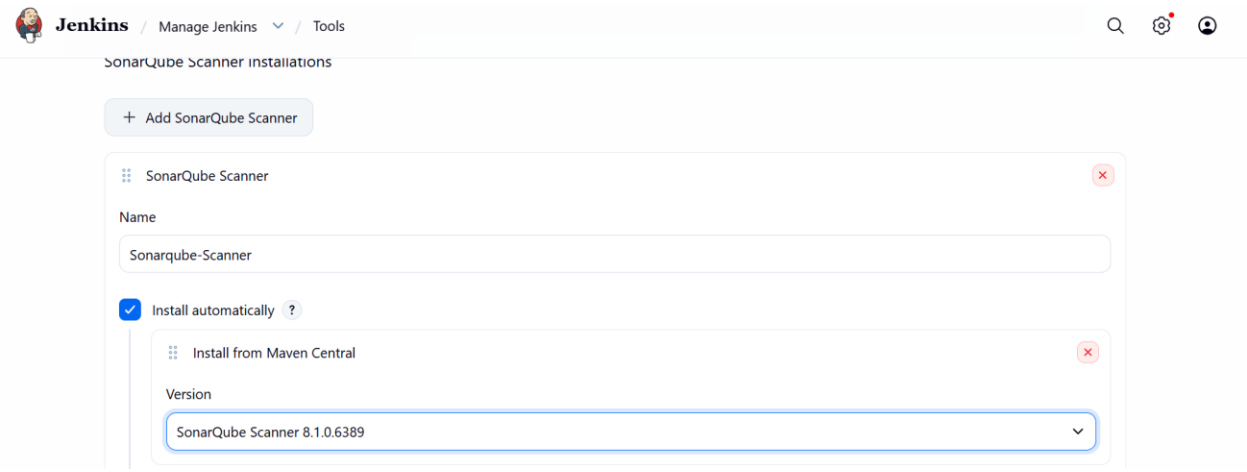


clone is successfully done

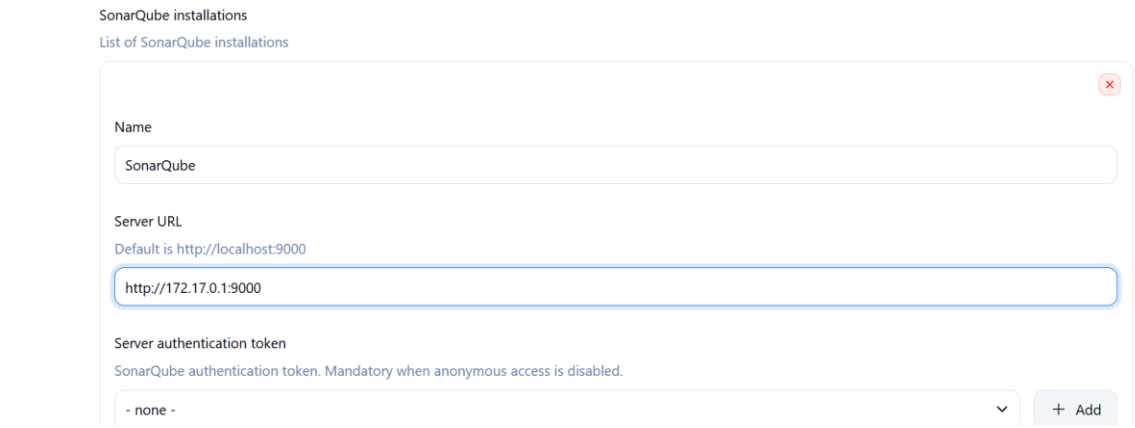
By allowing the port 9000



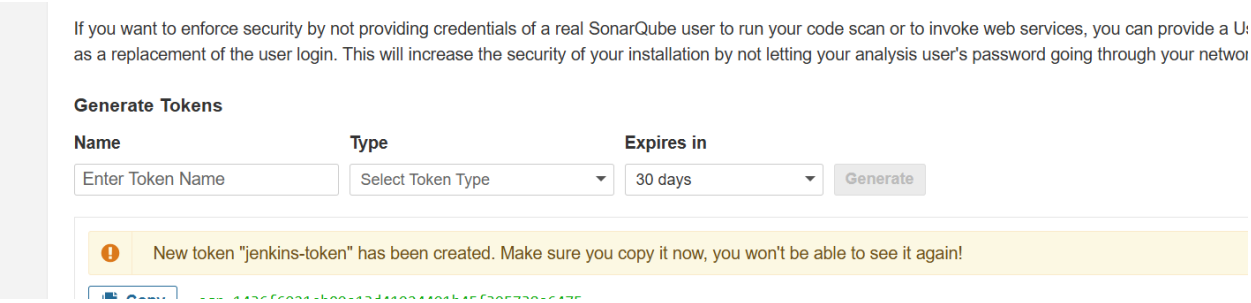
Install sonar qube scanner



Before putting the SonarQube command here, we need to tell Jenkins where the SonarQube server is running.



Then build now, got error due to no token



Generate token in sonarqube

Server authentication token

SonarQube authentication token. Mandatory when anonymous access is disabled.

sonar token

+ Add

Advanced

```
16:51:09.460 ERROR
16:51:09.460 ERROR Re-run SonarScanner CLI using the -X switch to enable full debug logging.
WARN: Unable to locate 'report-task.txt' in the workspace. Did the SonarScanner succeed?
ERROR: SonarQube scanner exited with non-zero code: 2
Finished: FAILURE
```

Jenkins / Hiring-App-Freestyle / #15

Status

Changes

Console Output

Delete build '#15'

Git Build Data

Edit build information

#15 (Oct 4, 2026, 7:48:46 PM)

Add description

Keep this build forever

Started by user admin

Started 13 sec ago

Took 8.1 sec

Revision: 9405b107fdffdf68e69fe953ea1f9669784f9d6a

Repository: https://github.com/betawins/hiring-app

- refs/remotes/origin/main

No changes.

Jenkins / Manage Jenkins / Plugins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

sla

Install

Install	Name ↓	Released	Health
☑	Slack Notification 795.v4b_9705b_e6d47 slack Build Notifiers Integrates Jenkins with Slack, allows publishing build statuses, messages and files to Slack channels.	1 yr 2 mo ago	92
☐	Translation Assistance 1.16		

Credential ?

jenkins slack bot token

+ Add

Default channel / member id ?

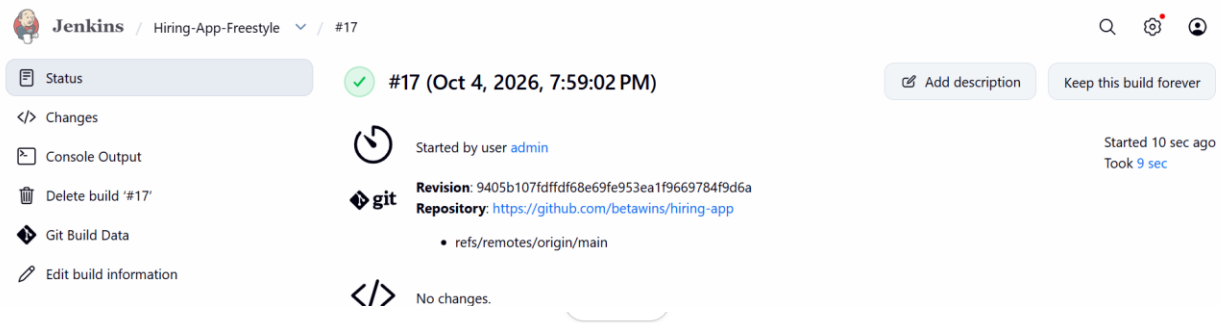
new-channel

☑ Custom slack app bot user ?

Advanced

Inserting Picture...

Test Connection



Jenkins Notifications APP 1:28 AM

Slack/Jenkins plugin: you're all set on http://98.92.204.165:8080/

Hiring-App-Freestyle - #17 Started by user admin (Open)

Hiring-App-Freestyle - #17 Success after 9 sec (Open)